

# Binary Codes

## Binary Codes

```
graph TD; A[Binary Codes] --> B[Numeric Codes]; A --> C[Alphanumeric Codes]; B --> B1[• 8421 Codes]; B --> B2[• 2421 Codes]; B --> B3[• Excess 3 Codes]; B --> B4[• Gray Codes]; C --> C1[• ASCII Codes]; C --> C2[• EBCDIC];
```

### Numeric Codes

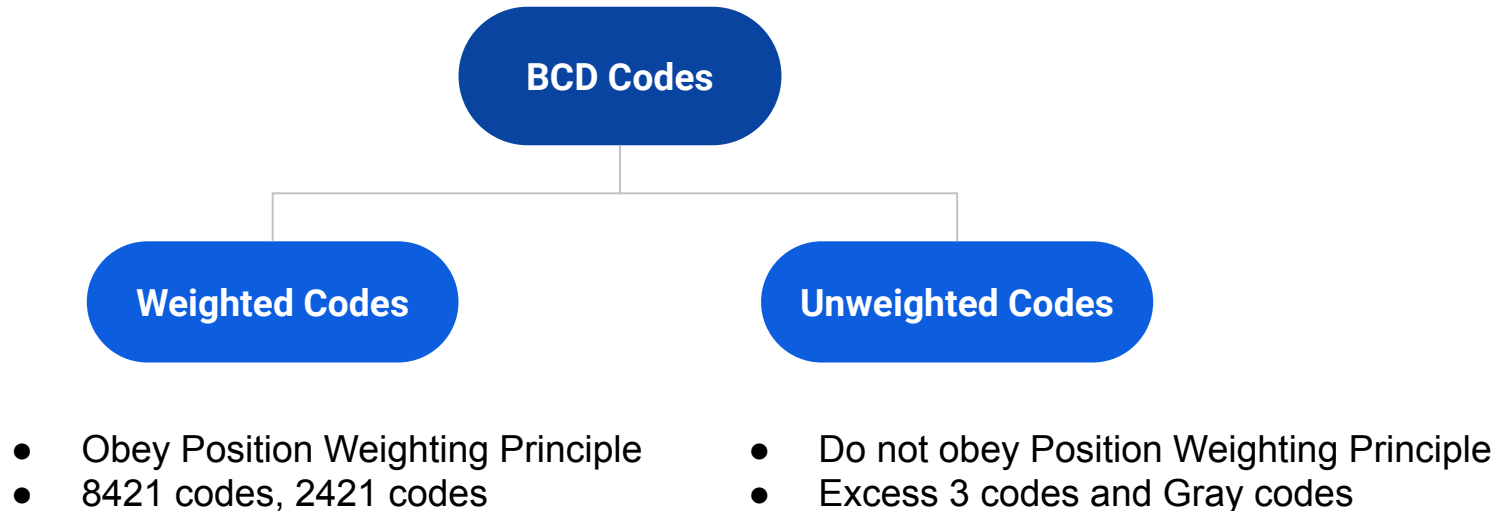
- 8421 Codes
- 2421 Codes
- Excess 3 Codes
- Gray Codes

### Alphanumeric Codes

- ASCII Codes
- EBCDIC

# Binary Coded Decimal (BCD) Codes

- Digits of a decimal number are encoded, one at a time, in group of 4 bits



- **Sequential Codes:** Succeeding code word is one binary number greater than its preceding code word
  - ❖ Mathematical manipulations easy
  - ❖ Eg: 8421, Excess 3
- **Self Complementing Codes:** Codeword of 9's complement of N, i.e.  $9-N$ , can be obtained from the code word of N by interchanging all the 0s and 1s
  - ❖ Eg: 2421, Excess 3
  - ❖ 8421 is not a self complementing code
- **Cyclic Codes (or) Unit distance codes:** Successive code word differs from the previous one in only 1 bit position
  - ❖ Eg: Cyclic Codes

| Decimal digit | 8 | 4 | 2 | 1 |
|---------------|---|---|---|---|
| 0             | 0 | 0 | 0 | 0 |
| 1             | 0 | 0 | 0 | 1 |
| 2             | 0 | 0 | 1 | 0 |
| 3             | 0 | 0 | 1 | 1 |
| 4             | 0 | 1 | 0 | 0 |
| 5             | 0 | 1 | 0 | 1 |
| 6             | 0 | 1 | 1 | 0 |
| 7             | 0 | 1 | 1 | 1 |
| 8             | 1 | 0 | 0 | 0 |
| 9             | 1 | 0 | 0 | 1 |

| 2 | 4 | 2 | 1 |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 |

| XS-3 |   |   |   |
|------|---|---|---|
| 0    | 0 | 1 | 1 |
| 0    | 1 | 0 | 0 |
| 0    | 1 | 0 | 1 |
| 0    | 1 | 1 | 0 |
| 0    | 1 | 1 | 1 |
| 1    | 0 | 0 | 0 |
| 1    | 0 | 0 | 1 |
| 1    | 0 | 1 | 0 |
| 1    | 0 | 1 | 1 |
| 1    | 1 | 0 | 0 |

# 8421 BCD Codes

| Decimal | BCD     |
|---------|---------|
|         | 8 4 2 1 |
| 0       | 0 0 0 0 |
| 1       | 0 0 0 1 |
| 2       | 0 0 1 0 |
| 3       | 0 0 1 1 |
| 4       | 0 1 0 0 |
| 5       | 0 1 0 1 |
| 6       | 0 1 1 0 |
| 7       | 0 1 1 1 |
| 8       | 1 0 0 0 |
| 9       | 1 0 0 1 |
| 10      | 1 0 1 0 |
| 11      | 1 0 1 1 |
| 12      | 1 1 0 0 |
| 13      | 1 1 0 1 |
| 14      | 1 1 1 0 |
| 15      | 1 1 1 1 |

# 8421 BCD Code

- Also known as simply **BCD code** or **natural binary code**
- Weighted and sequential code
- Easy conversion between BCD and decimal
- More number of bits required than binary
- Arithmetic operations are more complex

# BCD Addition

- Result is valid BCD when 4-bit sum is less than or equal to 9.

```

0101
0010
-----
0111
  ↓
  7
    
```

```

0011
0110
-----
1001
  ↓
  9
    
```

Binary addition

```

A3A2A1A0
B3B2B1B0
-----

```

C<sub>3</sub> S<sub>3</sub>S<sub>2</sub>S<sub>1</sub>S<sub>0</sub>

- Result is not valid when 4-bit sum is more than 9. Addition of 6 gives valid BCD.

```

0100      1011
0111      0110
-----
1011      (000)1 0001
  ↓        ↓   ↓
Not valid BCD  1   1
    
```

```

1001      (000)1 0001
1000      0110
-----
(000)1 0001  (000)1 0111
  ↓   ↓        ↓   ↓
  1   1        1   7
    
```



**EXAMPLE 3.1** Perform the following decimal additions in the 8421 code.

(a)  $25 + 13$

(b)  $679.6 + 536.8$

**Solution**

(a) 
$$\begin{array}{r} 25 \\ + 13 \\ \hline 38 \end{array} \Rightarrow \begin{array}{r} 0010 \quad 0101 \\ + 0001 \quad 0011 \\ \hline 0011 \quad 1000 \end{array}$$

(25 in BCD)  
(13 in BCD)  
(No carry, no illegal code. So, this is the correct sum)

(b) 
$$\begin{array}{r} 679.6 \\ + 536.8 \\ \hline 1216.4 \end{array} \Rightarrow \begin{array}{r} 0110 \quad 0111 \quad 1001 \quad .0110 \\ + 0101 \quad 0011 \quad 0110 \quad .1000 \\ \hline 1011 \quad 1010 \quad 1111 \quad .1110 \\ + 0110 \quad + 0110 \quad + 0110 \quad + .0110 \\ \hline \textcircled{1} 0001 \quad \textcircled{1} 0000 \quad \textcircled{1} 0101 \quad \textcircled{1} .0100 \\ +1 \quad \textcircled{1} \quad +1 \quad \textcircled{1} \quad +1 \quad \textcircled{1} \quad +1 \quad \textcircled{1} \\ \hline 0001 \quad 0010 \quad 0001 \quad 0110 \quad .0100 \\ 1 \quad 2 \quad 1 \quad 6 \quad .4 \end{array}$$

(679.6 in BCD)  
(536.8 in BCD)  
(All are illegal codes)  
(Add 0110 to each)  
(Propagate carry)  
(Corrected sum)

# BCD Subtraction

**EXAMPLE 3.2** Perform the following decimal subtractions in the 8421 BCD code.

(a)  $38 - 15$

(b)  $206.7 - 147.8$

**Solution**

|     |                                                                |               |                                                                                                                                                                                                                                              |                                                                                                                       |
|-----|----------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| (a) | $\begin{array}{r} 38 \\ - 15 \\ \hline 23 \end{array}$         | $\Rightarrow$ | $\begin{array}{r} 0011 \quad 1000 \\ - 0001 \quad 0101 \\ \hline 0010 \quad 0011 \end{array}$                                                                                                                                                | <p>(38 in BCD)</p> <p>(15 in BCD)</p> <p>(No borrow. So this is the correct difference.)</p>                          |
| (b) | $\begin{array}{r} 206.7 \\ - 147.8 \\ \hline 58.9 \end{array}$ | $\Rightarrow$ | $\begin{array}{r} 0010 \quad 0000 \quad 0110 \quad .0111 \\ - 0001 \quad 0100 \quad 0111 \quad .1000 \\ \hline 0000 \quad 1011 \quad 1110 \quad .1111 \\ - 0110 \quad - 0110 \quad -.0110 \\ \hline 0101 \quad 1000 \quad .1001 \end{array}$ | <p>(206.7 in BCD)</p> <p>(147.8 in BCD)</p> <p>(Borrows are present, subtract 0110)</p> <p>(Corrected difference)</p> |

# BCD Subtraction by 9's Complement

**EXAMPLE 3.3** Perform the following decimal subtractions in BCD by the 9's complement method.

- (a)  $85 - 24$       (b)  $305.5 - 168.8$       (c)  $679.6 - 885.9$

**Solution**

(a) 
$$\begin{array}{r} 85 \\ - 24 \\ \hline 61 \end{array} \Rightarrow \begin{array}{r} 85 \\ + 75 \quad (\text{9's complement of } 24) \\ \hline \textcircled{1} 60 \\ \text{↵} + 1 \quad (\text{End around carry}) \\ \hline 61 \end{array}$$

$$\begin{array}{r} 85_{10} \\ + 75_{10} \end{array} \Rightarrow \begin{array}{r} 1000 \quad 0101 \quad (85 \text{ in BCD}) \\ + 0111 \quad 0101 \quad (9's \text{ complement of } 24 \text{ in BCD}) \\ \hline 1111 \quad 1010 \quad (\text{Both are illegal codes, add } 0110 \text{ to each}) \\ + 0110 \quad + 0110 \\ \hline \textcircled{1} 0101 \quad \textcircled{1} 0000 \quad (\text{Propagate carry}) \\ + 1 \quad \text{↵} \quad + 1 \quad \text{↵} \\ \hline 1 \quad 0110 \quad 0000 \\ \text{↵} \quad \quad \quad + 1 \quad (\text{End around carry}) \\ \hline 0110 \quad 0001 \quad (\text{Corrected difference}) \end{array}$$

$$\begin{array}{r}
 \text{(b)} \quad 305.5 \\
 - 168.8 \\
 \hline
 136.7
 \end{array}
 \Rightarrow
 \begin{array}{r}
 305.5 \\
 + 831.1 \quad (\text{9's complement of } 168.8) \\
 \hline
 \textcircled{1} 136.6
 \end{array}$$

$$\begin{array}{r}
 \Downarrow + 1 \quad (\text{End around carry}) \\
 \hline
 136.7
 \end{array}$$

$$\begin{array}{r}
 305.5_{10} \\
 + 831.1_{10} \\
 \hline
 \end{array}
 \Rightarrow
 \begin{array}{r}
 0011 \quad 0000 \quad 0101 \quad .0101 \quad (\text{305.5 in BCD}) \\
 + 1000 \quad 0011 \quad 0001 \quad .0001 \quad (\text{9's complement of } 168.8 \text{ in BCD}) \\
 \hline
 + 1011 \quad 0011 \quad 0110 \quad .0110 \quad (1011 \text{ is an illegal code, add } 0110) \\
 + 0110
 \end{array}$$

$$\begin{array}{r}
 \textcircled{1} 0001 \quad 0011 \quad 0110 \quad .0110 \\
 \Downarrow + 1 \quad (\text{End around carry}) \\
 \hline
 0001 \quad 0011 \quad 0110 \quad .0111 \quad (\text{Corrected difference})
 \end{array}$$



$$\begin{array}{rcl}
 \text{(c)} & 679.6 & \\
 & - 885.9 & \Rightarrow \\
 \hline
 & - 206.3 & + 114.0 \quad (9\text{'s complement of } 885.9) \\
 & & \hline
 & & 793.6 \quad (\text{No carry})
 \end{array}$$

No carry. So the result is negative and is in 9's complement form. So the result is -206.3.

$$\begin{array}{rcl}
 679.6_{10} & \Rightarrow & 0110 \ 0111 \ 1001 \ .0110 \quad (679.6 \text{ in BCD}) \\
 + 114.0_{10} & & + 0001 \ 0001 \ 0100 \ .0000 \quad (9\text{'s complement of } 885.9 \text{ in BCD}) \\
 \hline
 & & 0111 \ 1000 \ 1101 \ .0110 \quad (1101 \text{ is an illegal code, add } 0110) \\
 & & \quad \quad \quad + 0110 \\
 \hline
 & & 0111 \ 1000 \ 1001 \ .0110 \quad (\text{Propagate carry}) \\
 & & \quad \quad \quad + 1 \quad \hookrightarrow \\
 \hline
 & & 0111 \ 1001 \ 0011 \ .0110 \quad (\text{No carry; } 793.6 \text{ in BCD})
 \end{array}$$

There is no carry and, therefore, the result is negative and is in its 9's complement form. The 9's complement of 793.6 is 206.3. So the result is -206.3.

**EXAMPLE 3.4** Perform the following subtractions in 8421 code using the 10's complement method.

(a)  $342.7 - 108.9$

(b)  $206.4 - 507.6$

**Solution**

$$\begin{array}{r}
 \text{(a)} \quad 342.7 \\
 - 108.9 \\
 \hline
 233.8
 \end{array}
 \Rightarrow
 \begin{array}{r}
 342.7 \\
 + 891.1 \quad (\text{10's complement of } 108.9) \\
 \hline
 \textcircled{1} 233.8 \quad (\text{Ignore carry})
 \end{array}$$

$$\begin{array}{r}
 342.7 \\
 + 891.1 \\
 \hline
 \end{array}
 \Rightarrow
 \begin{array}{r}
 0011 \quad 0100 \quad 0010 \quad .0111 \quad (\text{342.7 in BCD}) \\
 + 1000 \quad 1001 \quad 0001 \quad .0001 \quad (\text{10's complement of } 108.9 \text{ in BCD}) \\
 \hline
 1011 \quad 1101 \quad 0011 \quad .1000 \quad (\text{1011 and 1101 are illegal codes, add 0110 to each.})
 \end{array}$$

$$\begin{array}{r}
 \phantom{+ 1} + 0110 \quad + 0110 \\
 \hline
 \textcircled{1} 0001 \quad \textcircled{1} 0011 \quad 0011 \quad .1000 \quad (\text{Propagate carry}) \\
 + 1 \quad + 1
 \end{array}$$

$$\begin{array}{r}
 \textcircled{1} 0010 \quad 0011 \quad 0011 \quad .1000 \quad (\text{Ignore carry}) \\
 \hline
 0010 \quad 0011 \quad 0011 \quad .1000 \quad (\text{Corrected difference})
 \end{array}$$

$$\begin{array}{rcl}
 \text{(b)} & 206.4 & \Rightarrow 206.4 \\
 & - 507.6 & + 492.4 \text{ (10's complement of 507.6)} \\
 \hline
 & - 301.2 & 698.8 \text{ (No carry)}
 \end{array}$$

No carry. So the answer is negative and is in 10's complement form. The 10's complement of 698.8 is -301.2.

$$\begin{array}{rcl}
 206.4 & \Rightarrow & 0010 \quad 0000 \quad 0110 \quad .0100 \text{ (206.4 in BCD)} \\
 + 492.4 & + & 0100 \quad 1001 \quad 0010 \quad .0100 \text{ (10's complement of 507.6 in BCD)} \\
 \hline
 & & 0110 \quad 1001 \quad 1000 \quad .1000 \text{ (No illegal codes, no carry)}
 \end{array}$$

As there is no carry, the result is negative and is in its 10's complement form. The 10's complement of 698.8 is 301.2. So the corrected difference is -301.2.

# Logic for Addition of 6

| Decimal result | $C_3$ | $S_3$ | $S_2$ | $S_1$ | $S_0$ | $Y$<br>(Add 6) |
|----------------|-------|-------|-------|-------|-------|----------------|
| 0 - 9          | --    | --    | --    | --    | --    | 0              |
| 10             | 0     | 1     | 0     | 1     | 0     | 1              |
| 11             | 0     | 1     | 0     | 1     | 1     | 1              |
| 12             | 0     | 1     | 1     | 0     | 0     | 1              |
| 13             | 0     | 1     | 1     | 0     | 1     | 1              |
| 14             | 0     | 1     | 1     | 1     | 0     | 1              |
| 15             | 0     | 1     | 1     | 1     | 1     | 1              |
| 16             | 1     | 0     | 0     | 0     | 0     | 1              |
| 17             | 1     | 0     | 0     | 0     | 1     | 1              |
| 18             | 1     | 0     | 0     | 1     | 0     | 1              |

BCD Addition result is between 0 and 18.

$Y = 1$  when Addition of 6 is needed

$Y = 1$  when addition result is between 10 and 18.

i.e.  $C_3S_3S_2S_1S_0$  : 01010 to 10010

$Y = 0$  when  $C_3S_3S_2S_1S_0$  : 00000 to 01001

$Y = X$  (don't care) when  $C_3S_3S_2S_1S_0 > 10010$

This Truth Table on simplification gives

$$Y = C_3 + S_3S_2 + S_3S_1$$



# BCD Adder Unit

```

  49  0100 1001
+ 58  0101 1000
-----
    1 0001
    0110
-----
  1010 0111
  0110
-----
(000)1 0000 0111
  ↓   ↓   ↓
  1   0   7
  
```

